Computational Finance – Exercise Set 2

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Exercise 1

Consider the GBM model with the time-dependent function:

$$\begin{aligned} \mathrm{d}S(t) &= r \cdot S(t)\ \mathrm{d}t + \sigma(t) \cdot S(t)\ \mathrm{d}W^{\mathbb{Q}}(t) \\ \sigma(t) &= 0.6 - 0.2 \cdot e^{-1.5 \cdot t} \end{aligned}, \tag{1}$$

where $S(t_0) = 1$, $r = 0.05$ and the money-savings account is $\mathrm{d}M(t) = r \cdot M(t)\ \mathrm{d}t$, $M(t_0) = 1$.

a). *For the given model, derive Put Option prices analytically.*

The goal is to derive a closed-form expression for the European put option price $V_p(t_0, S_0)$ with strike price K and maturity time T.

First, the Feynman-Kac formula (Theorem 3.2.1. in Oosterlee and Grzelak (2019)) gives us the following pricing formula:

$$V_p(t_0, S_0) = e^{-r \cdot (T - t_0)} \cdot \mathbb{E}^{\mathbb{Q}} \left[\max\{K - S(T), 0\} \mid \mathcal{F}(t_0) \right]. \tag{2}$$

In order to derive this expression, we need to know the distribution of $S(T)$ under $\mathbb{Q}$.

Second, we apply a $\log(\cdot)$-transformation to $S(t)$ to obtain $X(t) := \log(S(t)) =: g(t, S(t))$. Since the following derivatives are continuous:

$$\begin{aligned} \frac{\partial g(t, S(t))}{\partial t} &= 0; \\ \frac{\partial g(t, S(t))}{\partial S(t)} &= \frac{1}{S(t)};, \\ \frac{\partial^2 g(t, S(t))}{\partial S^2(t)} &= \frac{-1}{S^2(t)} \end{aligned} \tag{3}$$

we can apply Itô's lemma (Theorem 2.1.1. in Oosterlee and Grzelak (2019)) to obtain the dynamics of $X(t)$:

$$\begin{aligned} \mathrm{d}g(t, S(t)) = \mathrm{d}X(t) &= \left(\frac{\partial g(t, S(t))}{\partial t} + \bar{\mu} \cdot \frac{\partial g(t, S(t))}{\partial S(t)} + \frac{1}{2} \cdot \bar{\sigma}^2 \cdot \frac{\partial^2 g(t, S(t))}{\partial S^2(t)} \right) \mathrm{d}t \\ &\quad + \bar{\sigma} \cdot \frac{\partial g(t, S(t))}{\partial S(t)}\ \mathrm{d}W^{\mathbb{Q}}(t) \\ &= \left[0 + r \cdot S(t) \cdot \frac{1}{S(t)} + \frac{1}{2} \frac{-1}{S^2(t)} \cdot \sigma^2(t) \cdot S^2(t) \right] \mathrm{d}t + \sigma(t) \cdot S(t) \cdot \frac{1}{S(t)}\ \mathrm{d}W^{\mathbb{Q}}(t) \\ &= \left[r - \frac{1}{2} \cdot \sigma^2(t) \right] \mathrm{d}t + \sigma(t)\ \mathrm{d}W^{\mathbb{Q}}(t). \end{aligned} \tag{4}$$

Third, since the RHS of the equation above does not depend on $X(t)$, we can integrate the SDE:

$$\begin{aligned}\int_{t_0}^{T} \mathrm{d}X(t) = X(T) - X(t_0) &= \int_{t_0}^{T} \left[r - \frac{1}{2} \cdot \sigma^2(t)\right] \mathrm{d}t + \int_{t_0}^{T} \sigma(t)\ \mathrm{d}W^{\mathbb{Q}}(t) \\ &\iff \\ X(T) = X(t_0) + \int_{t_0}^{T} &\left[r - \frac{1}{2} \cdot \sigma^2(t)\right] \mathrm{d}t + \int_{t_0}^{T} \sigma(t)\ \mathrm{d}W^{\mathbb{Q}}(t).\end{aligned} \tag{5}$$

Recall that we are interested in the distribution of $S(T)$. Since $X(T) = \log(S(T))$ is a function of $S(T)$, we are going to determine the distribution of $X(T)$ first. Notice that since independent increments of Brownian motion are normally distributed and $\sigma(t)$ is not stochastic, we know that $\int_{t_0}^{T} \sigma(t)\ \mathrm{d}W^{\mathbb{Q}}(t)$ follows a normal distribution (because this integral can be expressed as the limit of a summation, and the normal distribution is preserved under summation). Hence, $\int_{t_0}^{T} \sigma(t)\ \mathrm{d}W^{\mathbb{Q}}(t)$ is completely determined by its first two moments.
Using the property that the expectation of an Itô integral is zero results in:

$$\begin{aligned}\mathbb{E}[X(T)] &= X(t_0) + \int_{t_0}^{T} \left[r - \frac{1}{2} \cdot \sigma^2(t)\right] \mathrm{d}t + \mathbb{E}\left[\int_{t_0}^{T} \sigma(t)\ \mathrm{d}W^{\mathbb{Q}}(t)\right] \\ &= X(t_0) + \int_{t_0}^{T} \left[r - \frac{1}{2} \cdot \sigma^2(t)\right] \mathrm{d}t.\end{aligned} \tag{6}$$

Next, using equations (5) and (6), the Itô isometry property, and the fact that $\sigma(t)$ is not stochastic results in the following expression for the variance of $X(T)$:

$$\begin{aligned}\mathrm{Var}(X(T)) &= \mathbb{E}\left[(X(T) - \mathbb{E}[X(T)])^2\right] \\ &= \mathbb{E}\left[\left(\int_{t_0}^{T} \sigma(t)\ \mathrm{d}W^{\mathbb{Q}}(t)\right)^2\right] \\ &= \int_{t_0}^{T} \mathbb{E}[\sigma^2(t)]\ \mathrm{d}t = \int_{t_0}^{T} \sigma^2(t)\ \mathrm{d}t =: \tilde{\sigma}^2.\end{aligned} \tag{7}$$

Computing the integral $\tilde{\sigma}^2$:

$$\begin{aligned}\tilde{\sigma}^2 &:= \int_{t_0}^{T} (0.6 - 0.2 \cdot e^{-1.5 \cdot t})^2\ \mathrm{d}t \\ &= \int_{t_0}^{T} \left(0.6^2 - 2 \cdot 0.2 \cdot 0.6 \cdot e^{-1.5 \cdot t} + 0.2^2 \cdot e^{-3 \cdot t}\right)^2\ \mathrm{d}t \\ &= \int_{t_0}^{T} \left(0.36 - 0.24 \cdot e^{-1.5 \cdot t} + 0.04 \cdot e^{-3 \cdot t}\right)^2\ \mathrm{d}t \\ &= 0.36 \cdot t + \frac{0.24}{1.5} \cdot e^{-1.5 \cdot t} - \frac{0.04}{3} \cdot e^{-3 \cdot t}\Big|_{t_0}^{T} \\ &= 0.36 \cdot (T - t_0) + 0.16 \cdot (e^{-1.5 \cdot T} - e^{-1.5 \cdot t_0}) - \frac{0.04}{3} \cdot (e^{-3 \cdot T} - e^{-3 \cdot t_0})\end{aligned} \tag{8}$$

Therefore,

$$X(T) \sim \mathcal{N}\left(\mathbb{E}[X(T)],\ \tilde{\sigma}^2\right). \tag{9}$$

If we can rewrite the distribution of $X(T)$ in the following form with constant volatility σ_* (page 35 of Oosterlee and Grzelak (2019)):

$$X(T) \sim \mathcal{N}\left(X(t_0) + (\mu - \frac{1}{2}\sigma_*^2) \cdot (T - t_0), \quad \tilde{\sigma}^2 \cdot (T - t_0)\right), \tag{10}$$

then, since $X(T)$ is normally distribution ($\iff$ $S(T)$ is log-normally distributed), we can apply the Black-Scholes formula (Theorem 3.2.2. in Oosterlee and Grzelak (2019)) to obtain $V_p(t_0, S_0)$. Hence, the following equations need to be satisfied:

$$\begin{aligned} &(\mu - \frac{1}{2}\sigma_*^2) \cdot (T - t_0) = \int_{t_0}^{T} \left[r - \frac{1}{2} \cdot \sigma^2(t)\right] \mathrm{d}t = r \cdot (T - t_0) - \frac{1}{2}\tilde{\sigma}^2 \\ &\tilde{\sigma}^2 \cdot (T - t_0) = \tilde{\sigma}^2 \end{aligned}. \tag{11}$$

These equations are satisfied for:

$$\sigma_* := \sqrt{\frac{1}{T - t_0} \cdot \tilde{\sigma}^2}. \tag{12}$$

Note that $\mu = r$ under the $\mathbb{Q}$-measure.

So, by the Black-Scholes formula (Theorem 3.2.2. in Oosterlee and Grzelak (2019)), the price of the European put option is given by:

$$V_p(S_0, t_0) = K \cdot e^{-r \cdot (T - t_0)} \cdot \phi(-d_2(t_0, S_0)) - S_0 \cdot \phi(-d_1(t_0, S_0)), \tag{13}$$

with:

$$\begin{aligned} d_1(t_0, S_0) &= \frac{1}{\sigma_* \cdot \sqrt{T - t_0}} \cdot \left(\log\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma_*^2\right) \cdot (T - t_0)\right); \\ d_2(t_0, S_0) &= d_1(t_0, S_0) - \sigma_* \cdot \sqrt{T - t_o}; \\ \sigma_* &= \sqrt{\frac{1}{T - t_0} \cdot \left(0.36 \cdot (T - t_0) + 0.16 \cdot (e^{-1.5 \cdot T} - e^{-1.5 \cdot t_0}) - \frac{0.04}{3} \cdot (e^{-3 \cdot T} - e^{-3 \cdot t_0})\right)}. \end{aligned} \tag{14}$$

$\phi(\cdot)$ denotes the cumulative distribution function of a standard normal random variable.

b). *Perform a numerical experiment where you analyze the implied volatility smile/skew. Consider the implied volatility evolution for different expires. You can choose the settings for this numerical experiment freely. Discuss the results.*

We obtain the (Black Scholes) implied volatility by first computing the analytically derived European put option price formula in equation (13). Second, since the only (unknown) variable of the Black-Scholes (BS) formula is σ and we have a price for the put option, we solve the following root-finding problem by the Newton-Raphson algorithm (for σ_{imp}):

$$g(\sigma_{imp}) := V_p^{model}(K,T) - V_p(t_0, S_0; K, T, \sigma_{imp}, r) = 0. \tag{15}$$

$V_p^{model}(K,T)$ can be obtained by computing equation (13) with some strike K and some maturity time T, and $V_p(t_0, S_0; K, T, \sigma_{imp}, r)$ is essentially a function of σ_{imp} as it is the only unknown.

We set $t_0 := 0$, use $T \in \{1, 2, 3, 7, 10\}$ for the maturity times, and use `np.linspace(0.5, 2.5, 50)` as the grid of strike prices K.

The Newton-Raphson iterative process is as follows:

$$\sigma_{imp}^{(k+1)} = \sigma_{imp}^{(k)} - \frac{g\left(\sigma_{imp}^{(k)}\right)}{g'\left(\sigma_{imp}^{(k)}\right)}, \quad k \in \mathbb{N} \cup \{0\}, \tag{16}$$

where the denominator, called vega, is given by:

$$g'(\sigma) = -\frac{\partial V_p(t_0, S_0; K, T, \sigma, r)}{\partial \sigma} = -K \cdot e^{-r\cdot(T-t_0)} \phi(d_2(t_0, S_0)) \cdot \sqrt{T - t_0}. \tag{17}$$

Figure 1 depicts the graphs of the (approximated) implied volatility and the given time-dependent volatility $\sigma(t)$ on the $[0, 15]$ interval of maturity times. The implied volatility curve lies below the curve of $\sigma(t)$.

Furthermore, Figure 2 illustrates the implied volatility graphs over an interval of strike prices $K \in [0.50, 2.50]$, for five different maturity times T. All graphs are straight lines with a slope of zero, they are only translated vertically for different maturity times. The implied volatility graphs are straight lines because the European put option prices are modeled by the Black-Scholes formula which only allows for constant volatilities.

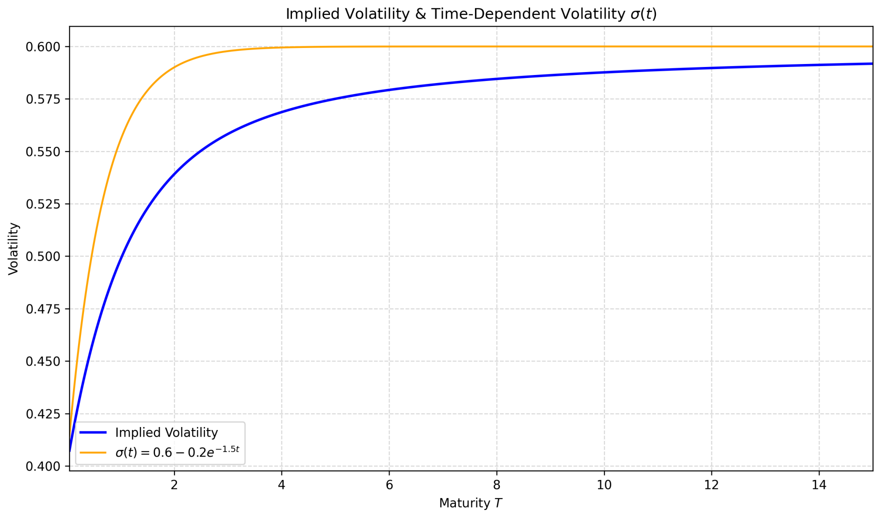

Figure 1: The graphs of the (approximated) implied volatility (blue) and the given time-dependent volatility $\sigma(t)$ (orange) on $[0, 15]$.

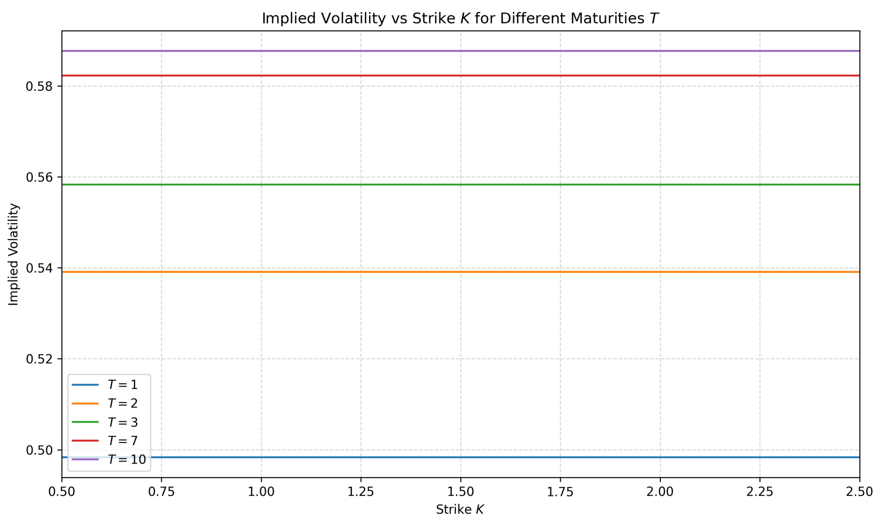

Figure 2: The (approximated) implied volatilities over a grid of strike prices K on $[0.50, 2.50]$.

Exercise 2

Under the assumption of "zero interest rates", $r = 0.0$*, consider the following model for the stock,* $S(t)$*,*

$$\mathrm{d}S(t) = \gamma \cdot \sqrt{S(t)}\ \mathrm{d}W(t), \quad S(t_0) > 0. \tag{18}$$

a). *Derive the corresponding characteristic function (CF),* $\phi_{S(T)}(u)$*, for* $S(T)$*.*

The goal is to find the CF $\phi_{S(T)}(u) = \mathbb{E}[e^{iuS(T)}|\mathcal{F}(t_0)]$.

First, we show that the SDE in equation (18) belongs to the class of affine diffusion (AD) processes, i.e., it must satisfy:

$$\mathrm{d}X(t) = \bar{\mu}(X(t))\ \mathrm{d}t + \bar{\sigma}(X(t)) \cdot Xt)\ \mathrm{d}W(t), \tag{19}$$

with the following affine (i.e., linear) functions:

$$\begin{aligned} \bar{\mu}(X(t)) &= a_0 + a_1 \cdot X(t), \quad a_0, a_1 \in \mathbb{R}; \\ \bar{\sigma}^2(X(t)) &= c_0 + c_1 \cdot X(t), \quad c_0, c_1 \in \mathbb{R}; \\ r(X(t)) &= r_0 + r_1 \cdot X(t), \quad r_0, r_1 \in \mathbb{R}. \end{aligned} \tag{20}$$

In our model, $X(t) = S(t)$:

- $\bar{\mu}(S(t)) = 0 \implies a_0 = a_1 := 0$;
- $\bar{\sigma}(S(t)) = \gamma \cdot \sqrt{S(t)} \implies \bar{\sigma}^2(S(t)) = \gamma^2 \cdot S(t) \implies c_0 := 0$ and $c_1 := \gamma^2$;
- $r(S(t)) = 0 \implies r_0 = r_1 := 0$.

Hence, the SDE in equation (18) belongs to the AD class.

Second, we use the result by Duffie, Pan and Singleton (2000) that for AD processes, the discounted CF defined as:

$$\phi_{S(T)}(S(t), t, T, u) = \mathbb{E}\left[e^{-\int_{t_0}^{T} r(S(v)\ \mathrm{d}v} \cdot e^{iuS(T)}|\mathcal{F}(t_0)\right] = \mathbb{E}\left[e^{iuS(T)}|\mathcal{F}(t_0)\right], \tag{21}$$

with boundary condition:

$$\phi_{S(T)}(S(T), T, T, u) = e^{iuS(T)}, \tag{22}$$

has a solution of the following form:

$$\phi_{S(T)}(S(T), t, T, u) = e^{A(u,t,T) + B(u,t,T) \cdot S(T)}. \tag{23}$$

Often, we simplify the notation of $\phi_{S(T)}(S(T), t, T, u)$ to $\phi_{S(T)}(u)$. Furthermore, the boundary conditions can be equivalently stated as:

$$\begin{aligned} A(u, T, T) &\equiv A(u, \tau) = 0; \\ B(u, T, T) &\equiv B(u, \tau) = iu, \end{aligned} \tag{24}$$

where $\tau := T - t_0$.

Third, we determine $A(u, \tau)$ and $B(u, \tau)$ by using the fact that they have to satisfy the following systems of Riccati-type ODEs:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\tau} A(u, \tau) &= -r_0 + a_0 \cdot B(u, \tau) + \frac{1}{2} \cdot c_0 \cdot B^2(u, \tau) = 0 + 0 + 0 = 0; \\ \frac{\mathrm{d}}{\mathrm{d}\tau} B(u, \tau) &= -r_1 + a_1 \cdot B(u, \tau) + \frac{1}{2} \cdot c_1 \cdot B^2(u, \tau) = 0 + 0 + \frac{1}{2} \cdot \gamma^2 \cdot B^2(u, \tau) = \frac{1}{2} \cdot \gamma^2 \cdot B^2(u, \tau). \end{aligned} \tag{25}$$

Hence, we need to solve these ODEs by using the boundary conditions in equation (24). The ODE for $A(u,\tau)$ can be trivially solved:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}A(u,\tau) = 0 \quad \& \quad A(u,0) = 0 \quad \Longrightarrow \quad A(u,\tau) = 0. \tag{26}$$

Next:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}B(u,\tau) = \frac{1}{2}\cdot\gamma^2\cdot B^2(u,\tau) \quad \& \quad B(u,0) = iu. \tag{27}$$

Solving the ODE by separation of variables yields:

$$\begin{aligned} \frac{\mathrm{d}B(u,\tau)}{B^2(u,\tau)} &= \frac{1}{2}\cdot\gamma^2 \ \mathrm{d}\tau \\ \int \frac{\mathrm{d}B(u,\tau)}{B^2(u,\tau)} &= \int \frac{1}{2}\cdot\gamma^2 \ \mathrm{d}\tau \\ \frac{-1}{B(u,\tau)} &= \frac{1}{2}\cdot\gamma^2\cdot\tau + C, \quad C \in \mathbb{R}. \end{aligned} \tag{28}$$

Using the boundary condition to find C:

$$\frac{-1}{iu} = \frac{1}{2}\cdot\gamma^2\cdot 0 + C \quad \Longrightarrow \quad C = \frac{-1}{iu} = \frac{-1}{i}\cdot\frac{1}{u} = \frac{i}{u}. \tag{29}$$

Hence:

$$B = \frac{-1}{\frac{1}{2}\cdot\gamma^2\cdot\tau + \frac{i}{u}} = \frac{-u}{\frac{1}{2}\cdot\gamma^2\cdot u\cdot\tau + i}. \tag{30}$$

Checking whether the boundary condition still holds:

$$B(u,0) = \frac{-u}{i} = \frac{-1}{i}\cdot u = iu. \tag{31}$$

So, the CF of $S(T)$ is given by:

$$\phi_{S(T)}(u) = \exp\left(\frac{-u}{\frac{1}{2}\cdot\gamma^2\cdot u\cdot\tau + i}\cdot S(t_0)\right). \tag{32}$$

b). *Consider $\gamma = 0.25$, $S(t_0) = 1$ and $T = 5$. Using the COS method, recover the probability density function, PDF, of $S(T)$ and compare the results against the analytical solution (if you are unable to determine the exact PDF for this model, please consider a simple Monte Carlo simulation with Euler discretization).*

We consider a simple Monte Carlo (MC) simulation with Euler discretization instead of the analytical PDF. The SDE:

$$\mathrm{d}S(t) = \gamma \cdot \sqrt{S(t)}\, \mathrm{d}W(t) =: \bar{\sigma}(t, S(t))\, \mathrm{d}W(t), \tag{33}$$

with $\bar{\mu}(t, X(t)) = 0$, is discretized by the Euler scheme as follows:

$$\begin{aligned} s_{i+1,j} &= s_{i,j} + \bar{\sigma}(t_i, S(t_i)) \cdot \Delta W_{i,j} = s_{i,j} + \gamma \cdot \sqrt{S(t_i)} \cdot \Delta W_{i,j}; \\ \Delta W_{i,j} &= W_{i+1,j} - W_{i,j} = \sqrt{\Delta t} \cdot Z_j, \quad Z_j \sim \mathcal{N}(0,1); \\ t_i &= i \cdot \Delta t, \end{aligned} \tag{34}$$

where i denotes the i^{th} time(step) and j denotes the j^{th} path.

Moreover, the Fourier-cosine expansion of the PDF $f_{S(T)}$ on the truncated domain $[a, b] \subsetneq \mathbb{R}$ is given by:

$$f_{S(T)}(s) = {\sum_{k=0}^{\infty}}' F_k \cdot \cos\left(k \cdot \pi \cdot \frac{s-a}{b-a}\right), \tag{35}$$

with $s \in [a, b]$ and:

$$\begin{aligned} F_k &= \frac{2}{b-a} \int_a^b f_{S(T)}(s) \cdot \cos\left(k \cdot \pi \cdot \frac{s-a}{b-a}\right) \mathrm{d}s \\ &\approx \frac{2}{b-a} \cdot \mathrm{Re}\left(\phi_{S(T)}(\frac{k \cdot \pi}{b-a}) \cdot \exp\left(-i \cdot \frac{k \cdot a \cdot \pi}{b-a}\right)\right) =: \bar{F}_k. \end{aligned} \tag{36}$$

We approximate this expression of $f_{S(T)}$ by truncating the series summation, i.e., the COS method:

$$\hat{f}_{S(T)}(s) \approx {\sum_{k=0}^{N}}' \hat{F}_k \cdot \cos\left(k \cdot \pi \cdot \frac{s-a}{b-a}\right), \quad N \in \mathbb{N}. \tag{37}$$

Note that the prime in the summation means that the first term is scaled by 0.5.

Figures 3, 4 and 5 depict the Monte Carlo approximation of the PDF for different `NoOfPaths` and `NoOfSteps`, and seven approximations of the COS method for different N are presented. A (sufficiently) large number of `NoOfPaths` is more important than `NoOfSteps` for the convergence of the histogram representing the MC approximation. For the histograms, we use the Freedman-Diaconis rule for the optimal bin width choice:

$$h = 2 \cdot IQR\left(\{s_{m,j}\}_{j=1}^{n}\right) \cdot n^{-1/3}, \tag{38}$$

where h denotes the bin width, IQR denotes the interquartile range, and n the number of samples used. Furthermore, the COS method seems to converge well to the PDF $f_{S(T)}$ for $N \geq 16$, because higher values of N give almost indistinguishable PDF approximations by solely looking at the graphs. The COS method does not yield a good approximation for $N = 4$ and $N = 8$.

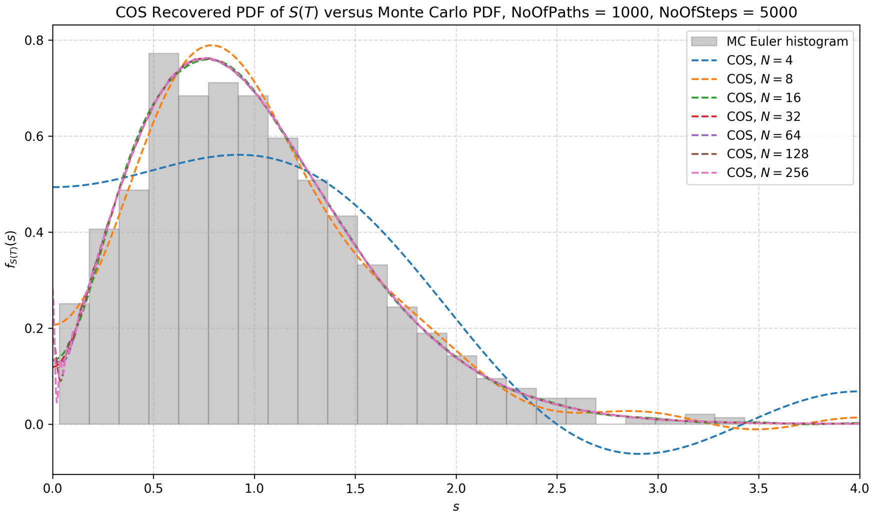

Figure 3: The Monte Carlo approximation of the PDF with `NoOfPaths`=1000 and `NoOfSteps`=5000, and seven approximations of the COS method for different N.

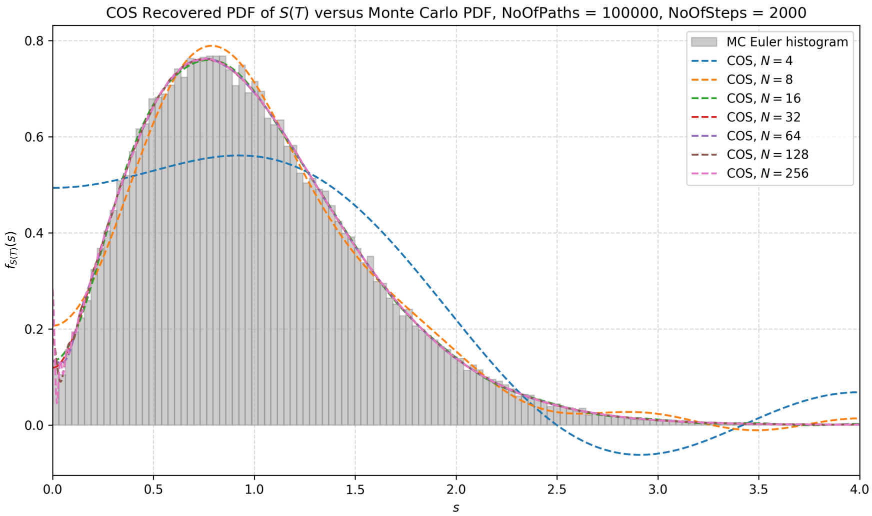

Figure 4: The Monte Carlo approximation of the PDF with `NoOfPaths`=100,000 and `NoOfSteps`=2000, and seven approximations of the COS method for different N.

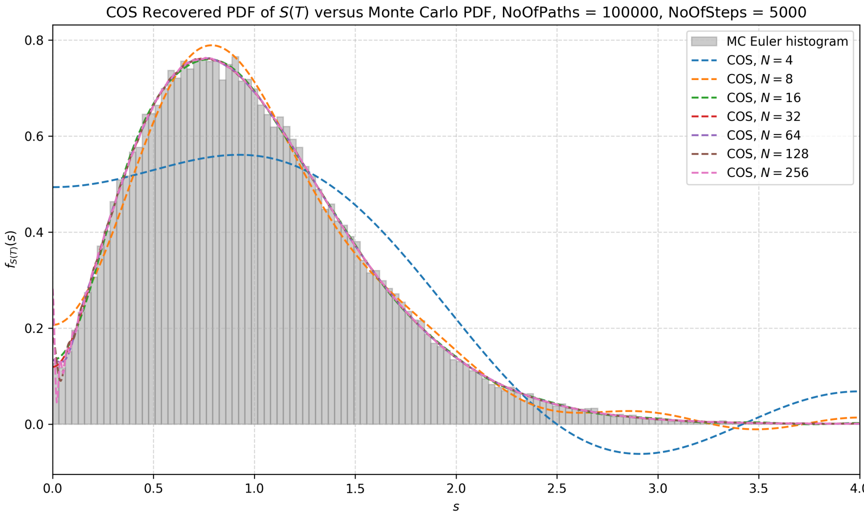

Figure 5: The Monte Carlo approximation of the PDF with `NoOfPaths`=100,000 and `NoOfSteps`=5000, and seven approximations of the COS method for different N.

Exercise 3

Perform a Monte Carlo simulation for a two-asset option with payoff $S_1(T)\cdot\max\{S_1(T), S_2(T)\}$, where the asset prices are governed by correlated Brownian motion, $\sigma_1 = 0.4$, $\sigma_2 = 0.15$, $r = 0.01$, $S_1(0) = S_2(0) = 1$. Consider in two numerical experiments $\rho = -0.9$ and $\rho = 0.9$, respectively. Give confidence intervals for the results, and describe the observations regarding positive and negative correlation.

Under the risk-neutral measure $\mathbb{Q}$, we consider the following SDEs for two correlated GBM assets:

$$\begin{aligned} \mathrm{d}S_1(t) &= r \cdot S_1(t)\ \mathrm{d}t + \sigma_1 \cdot S_1(t)\ \mathrm{d}W_1(t) \\ \mathrm{d}S_2(t) &= r \cdot S_2(t)\ \mathrm{d}t + \sigma_2 \cdot S_2(t)\ \mathrm{d}W_2(t) \end{aligned}, \tag{39}$$

with $\mathrm{d}W_1(t) \cdot \mathrm{d}W_2(t) = \rho \cdot \mathrm{d}t$, where ρ denotes the correlation between W_1 and W_2.

To simulate $S_i(t)$, $i \in \{1, 2\}$, we rewrite the system of correlated Brownian motions in terms of independent Brownian motions $\tilde{\mathbf{W}} = [\tilde{W}_1(t), \tilde{W}_2(t)]^T$ by using the Cholesky decomposition:

$$\begin{aligned} W_1(t) &:= \tilde{W}_1(t); \\ W_2(t) &:= \rho \cdot \tilde{W}_1(t) + \sqrt{1-\rho^2} \cdot \tilde{W}_2(t). \end{aligned} \tag{40}$$

Since $\mathrm{Cov}(W_1(t), W_2(t)) = \rho$, the correlation between $W_1(t)$ and $W_2(t)$ equals ρ as desired.

Equation (2.17) in Oosterlee and Grzelak (2019) presents a formula to obtain the exact value of $S_i(T)$, $i \in \{1, 2\}$:

$$\begin{aligned} S_i(T) &= S_i(t_0) \cdot \exp\left(\left(\mu - \frac{1}{2}\cdot\sigma_i^2\right)\cdot(T - t_0) + \sigma_i \cdot (W_i(T) - W_i(t_0))\right) \\ &= S_i(0) \cdot \exp\left(\left(r - \frac{1}{2}\cdot\sigma_i^2\right)\cdot T + \sigma_i \cdot W_i(T)\right), \end{aligned} \tag{41}$$

for $i \in \{1, 2\}$, and since $t_0 := 0$ and $\mu = r$ under the $\mathbb{Q}$-measure.

Moreover, using the Feynman-Kac formula (Theorem 3.2.1. in Oosterlee and Grzelak (2019)), we obtain a formula for the value of the two-asset option:

$$V(t_0) = e^{-r\cdot T} \cdot \mathbb{E}^{\mathbb{Q}}\left[S_1(T) \cdot \max\{S_1(T), S_2(T)\}\right]. \tag{42}$$

The expectation can be approximated by simple Monte Carlo (integration) yielding the following Monte Carlo approximation of $V(t_0)$:

$$\hat{V}(t_0) = e^{-r\cdot T} \cdot \frac{1}{N}\sum_{j=1}^{N} S_{1,j}(T) \cdot \max\{S_{1,j}(T), S_{2,j}(T)\}, \tag{43}$$

for sufficiently large N, where:

$$S_{i,j}(T) = S_{i,j}(0) \cdot \exp\left(\left(r - \frac{1}{2}\cdot\sigma_{i,j}^2\right)\cdot T + \sigma_i \cdot W_{i,j}(T)\right), \tag{44}$$

for $i \in \{1, 2\}$. Note that $W_1(t)$ and $W_2(t)$ are generated as follows:

$$\begin{aligned} W_{1,j}(T) &:= \sqrt{T} \cdot Z_{1,j}; \\ W_{2,j}(T) &:= \rho \cdot \sqrt{T} \cdot Z_{1,j} + \sqrt{1-\rho^2} \cdot \sqrt{T} \cdot Z_{2,j}. \end{aligned} \tag{45}$$

for independent $Z_{1,j}, Z_{2,j} \sim \mathcal{N}(0,1)$.

The standard error of $\hat{V}(t_0)$ is $\epsilon_N := \frac{1}{\sqrt{N}} \cdot \bar{v}_N$, where $\bar{v}_N$ denotes the sample variance. Hence, the 95% confidence interval of $\hat{V}$ is:

$$\left[\hat{V}(t_0) - 1.96 \cdot \epsilon_N, \hat{V}(t_0) + 1.96 \cdot \epsilon_N\right]. \tag{46}$$

The 95% confidence interval is $[1.243, 1.257]$ for the $\rho = -0.9$ case and $[1.303, 1.316]$ for the $\rho = 0.9$ case, both rounded to three decimals. Figure 6 shows that a negative correlation ($\rho = -0.9$) yields a higher payoff than a positive correlation ($\rho = 0.9$). Observe that both expected payoffs do not fall inside their confidence intervals, making the result not statistically significant. Similar to exercise 2, the Freedman-Diaconis rule is used in the creation of the histograms.

Figure 7 illustrates the relation between the two assets for the two correlation values. A negative correlation implies an opposite tendency, e.g., if one goes down the other goes up. And a positive correlation implies an equivalent tendency, e.g., if one goes up the other also goes up. This structure is present due to the right-left and left-right diagonal in the scatter plots, respectively.

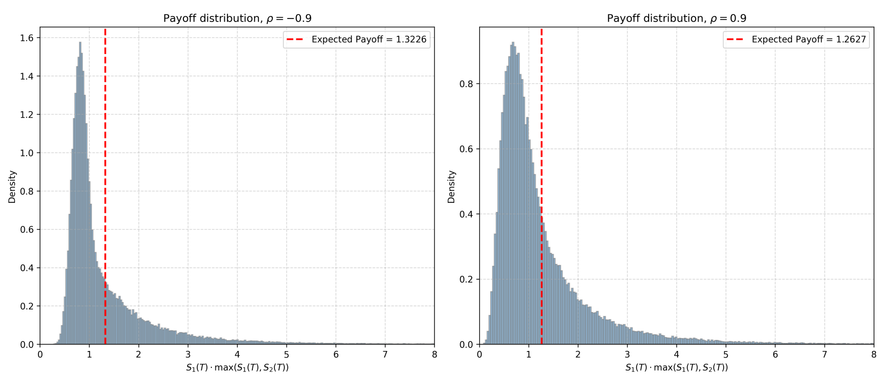

Figure 6

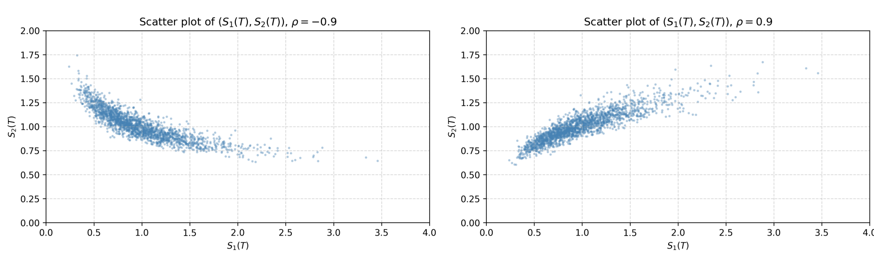

Figure 7

References

Oosterlee, C. W. and Grzelak, L. A. (2019). *Mathematical Modeling and Computation in Finance: With Exercises and Python and MATLAB Computer Codes*. World Scientific Publishing.